Mock 2 M02

Subscription Lifecycle - Delta Change Data Feed

SUBMISSION TEMPLATE

Difficulty	Marks	Tasks	Screenshot rule
Medium	20	5	Multiple allowed per task

Trainee: _____Soumiya Ayyappan_____ Candidate ID: ____10877547_____ Catalog: _____cat_ltm10877547

Enable Delta Change Data Feed before mutations, process subscription changes and reconcile change events against the final snapshot.

Evidence policy

SCREENSHOT POLICY: There is NO one-task/one-screenshot limit. For every assessed task, provide as many readable images as needed to establish ALL listed rubric criteria. Where the rubric tests implementation AND its effect, supply both code/configuration screenshots and result screenshots. If code is long, use consecutive clearly labelled Code A, Code B, Code C images; do not shrink or stitch until unreadable. Label images Q_T01_Code_A, Q_T01_Result_A, etc. Keep table/pipeline names, executed statements and relevant returned rows visible. An output screenshot cannot replace required implementation evidence, and code alone cannot replace required output evidence. Never rerun MERGE, DELETE or other mutations solely to capture evidence: capture prior run/history and current read-only validations.

Environment and object names

Use your existing assigned Unity Catalog and schema mock2. Use the approved existing external location; the instructor supplies your trainee-specific ADLS URI. Create a question-specific EXTERNAL volume at that full child URI. Do not create a new catalog, storage credential or external location. Upload original CSV filenames into input/ under the question volume. Store managed Delta tables and pipeline outputs in schema managed storage, not in the volume.

Setting	Required value
Catalog	Your assigned existing catalog
Schema	mock2
External volume	m2_m02_files
Child ADLS path	mock2/m02/files/ beneath your assigned trainee root
Input folder	/Volumes/<catalog>/mock2/m2_m02_files/input/
Required output objects	m2_m02_subscriptions, m2_m02_change_input

Key terms

- CDF: records row-level inserts, deletes and pre/post images of updates made after it is enabled
- preimage/postimage: old and new representations of an updated record
- baseline version: actual Delta history version immediately before the first assessed mutation

Source dataset and column dictionary

Provided file(s): M02_subscriptions_base.csv, M02_subscriptions_changes.csv

Column	Required type	Use in operations
subscription_id	STRING	Primary business identifier
customer_id	STRING	Customer
plan_name	STRING	Subscription plan
monthly_fee	INTEGER	Monthly charge
operation	STRING	Change-file-only action: UPDATE, DELETE, INSERT

Task requirements and evidence

Task 1. Prepare CDF-enabled baseline | 4 marks

1.1 Upload both original CSV files to the question EXTERNAL volume.

1.2 Load the six baseline subscriptions to a managed Delta target using the first four columns.

1.3 Enable Delta Change Data Feed before any assessed update/insert/delete; verify the CDF table property.

1.4 Record the actual table-history version immediately before the first mutation; do not assume it is zero.

Expected result: 6 baseline records; CDF property true; actual baseline version captured.

Evaluator evidence - every item below must be legible:

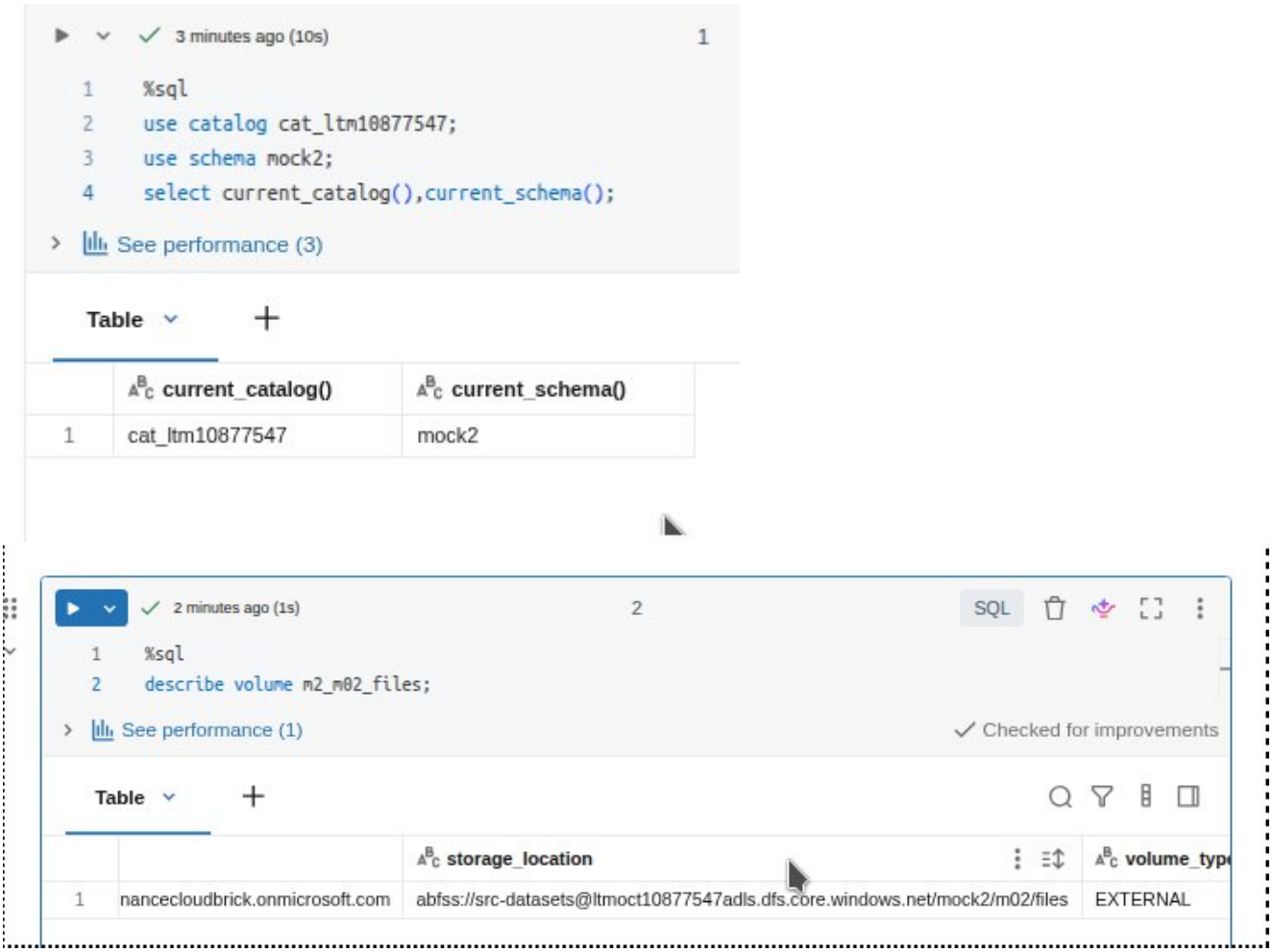
A. Setup: volume location and both filenames.

B. Code/config: baseline managed Delta table and CDF enablement.

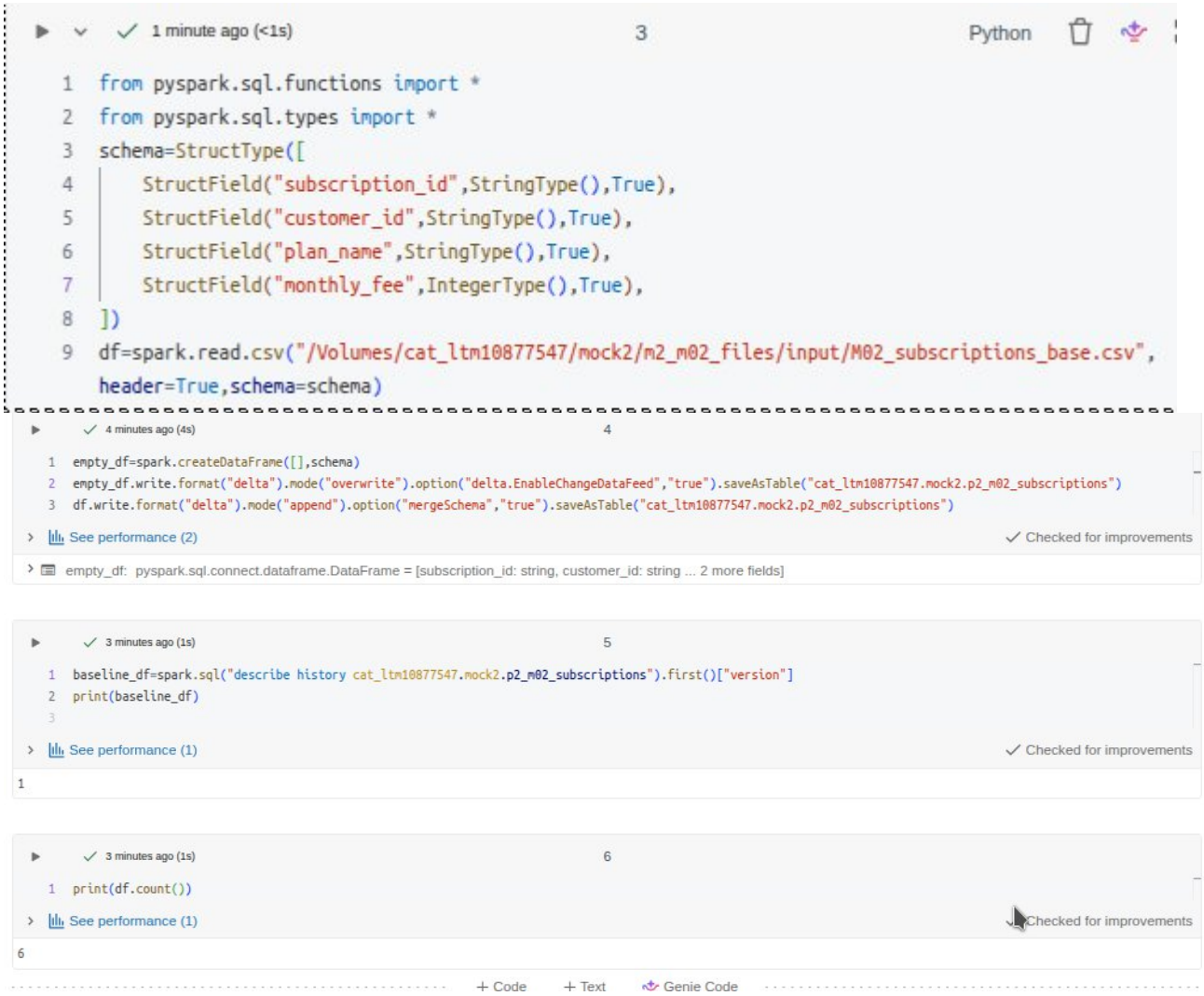
C. Results: CDF property true, initial row count 6 and baseline history version.

Use filenames such as M2_M02_T01_Code_A.png, M2_M02_T01_Result_A.png or M2_M02_T01_UI_A.png. Add _B / _C where required. These are naming examples, NOT a fixed screenshot count.

SCREENSHOT PLACEHOLDER - paste the required task evidence below; duplicate this holder for additional screenshots.



[Paste readable screenshot here]



Additional screenshots (Code B / Result B / UI B, etc.) may be inserted after this placeholder.

Evidence reference / workspace object: _____

Task 2. Prepare typed change instructions | 3 marks

2.1 Load all four change-file records into a separate table, keeping operation column.

2.2 Verify one UPDATE, one DELETE and two INSERT actions; ensure target matching is by subscription_id.

Expected result: UPDATE 1; DELETE 1; INSERT 2; four incoming records.

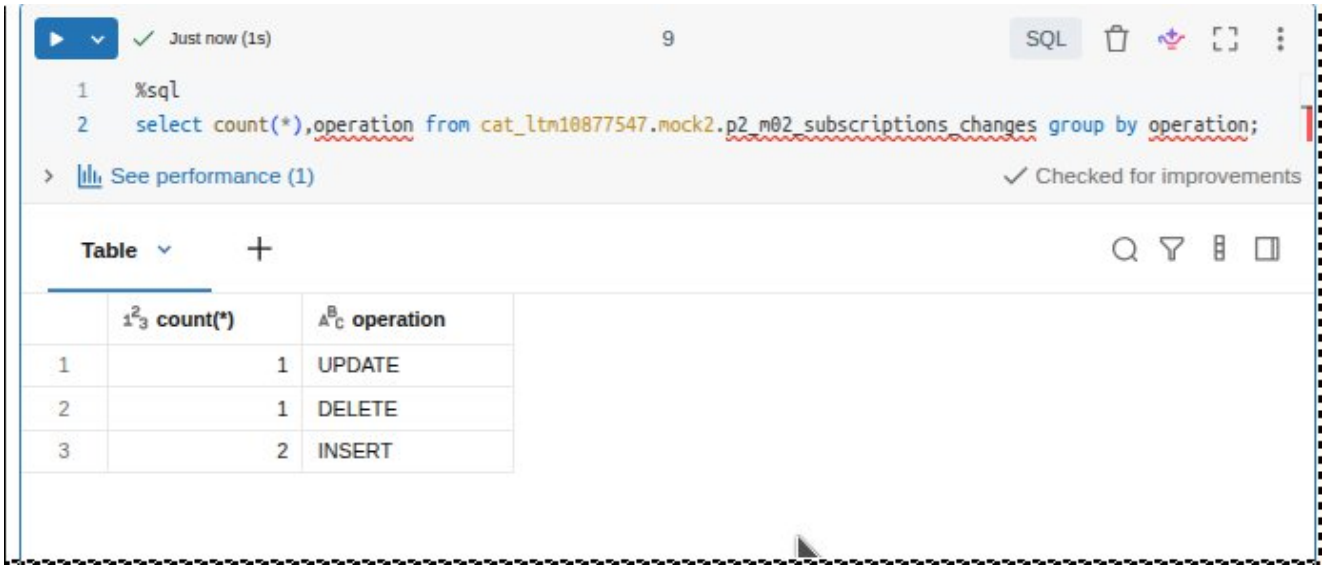
Evaluator evidence - every item below must be legible:

A. Code: typed change-file load with operation retained.

B. Result: grouped operation counts 1/1/2.

Use filenames such as M2_M02_T02_Code_A.png, M2_M02_T02_Result_A.png or M2_M02_T02_UI_A.png. Add _B / _C where required. These are naming examples, NOT a fixed screenshot count.

SCREENSHOT PLACEHOLDER - paste the required task evidence below; duplicate this holder for additional screenshots.



[Paste readable screenshot here]

Additional screenshots (Code B / Result B / UI B, etc.) may be inserted after this placeholder.

Evidence reference / workspace object: _____

Task 3. Apply updates and inserts | 5 marks

3.1 Apply the UPDATE and INSERT records to the baseline Delta target using a Delta MERGE.

3.2 Match subscription_id, update plan_name and monthly_fee for S102, and insert S107 and S108 with all four business fields.

3.3 Preserve unchanged subscriptions; do not process the DELETE action in this step.

Expected result: S102 Premium/999; S107 Basic/299; S108 Pro/599; snapshot after MERGE 8 rows.

Evaluator evidence - every item below must be legible:

A. Code A/B if necessary: complete MERGE join and matched/unmatched actions.

B. Results: three named affected records and 8-row snapshot.

C. History: actual successful MERGE version, if shown separately.

Use filenames such as M2_M02_T03_Code_A.png, M2_M02_T03_Result_A.png or M2_M02_T03_UI_A.png. Add _B / _C where required. These are naming examples, NOT a fixed screenshot count.

SCREENSHOT PLACEHOLDER - paste the required task evidence below; duplicate this holder for additional screenshots.

Table +

	subscription_id	plan_name	monthly_fee
1	S101	Basic	299
2	S103	Basic	299
3	S104	Premium	999
4	S105	Pro	599
5	S106	Basic	299
6	S108	Pro	599
7	S107	Basic	299
8	S102	Premium	999

[Paste readable screenshot here]

Additional screenshots (Code B / Result B / UI B, etc.) may be inserted after this placeholder.

Evidence reference / workspace object: _____

Task 4. Apply deletion and inspect CDF | 5 marks

4.1 Delete S104 using the operation instruction. Do not rerun the change transaction.

4.2 Read Change Data Feed beginning with the first change version AFTER the captured baseline version; include all assessed mutations.

4.3 Group change records by _change_type and compare with expected insert, delete, update_preimage, update_postimage event counts.

Expected result: insert 2; delete 1; update_preimage 1; update_postimage 1; five CDF events.

Evaluator evidence - every item below must be legible:

A. Code: deletion logic referencing S104 and operation DELETE.

B. Code: CDF read with correctly recorded starting version and grouping by change type.

C. Result: four event categories with counts 2,1,1,1; history/version evidence if not already shown.

Use filenames such as M2_M02_T04_Code_A.png, M2_M02_T04_Result_A.png or M2_M02_T04_UI_A.png. Add _B / _C where required. These are naming examples, NOT a fixed screenshot count.

SCREENSHOT PLACEHOLDER - paste the required task evidence below; duplicate this holder for additional screenshots.

Table

	subscription_id	customer_id	plan_name	monthly_
1	S102	C02	Premium	
2	S104	C04	Premium	
3	S107	C07	Basic	
4	S108	C08	Pro	

Just now (1s) 15 SQL

```
%sql
select count(*),operation from cat_ltm10877547.mock2.p2_m02_subscriptions_changes group by operation
```

See performance (1) Checked for improvements

Table

	count(*)	operation
1	1	UPDATE
2	1	DELETE
3	2	INSERT

[Paste readable screenshot here]

Additional screenshots (Code B / Result B / UI B, etc.) may be inserted after this placeholder.

Evidence reference / workspace object: _____

Task 5. Check final snapshot and event reconciliation | 3 marks

5.1 Show final Delta target contents and verify S104 is absent and S102 is revised.

5.2 Check final count of seven rows; reconcile baseline 6 + inserts 2 - deletes 1 = 7.

5.3 Show table history versions and confirm all assessed changes occur after CDF was enabled.

Expected result: 7 subscriptions; S104 absent; S102 Premium/999; CDF counts as above.

Evaluator evidence - every item below must be legible:

A. Code: final snapshot and row-count reconciliation.

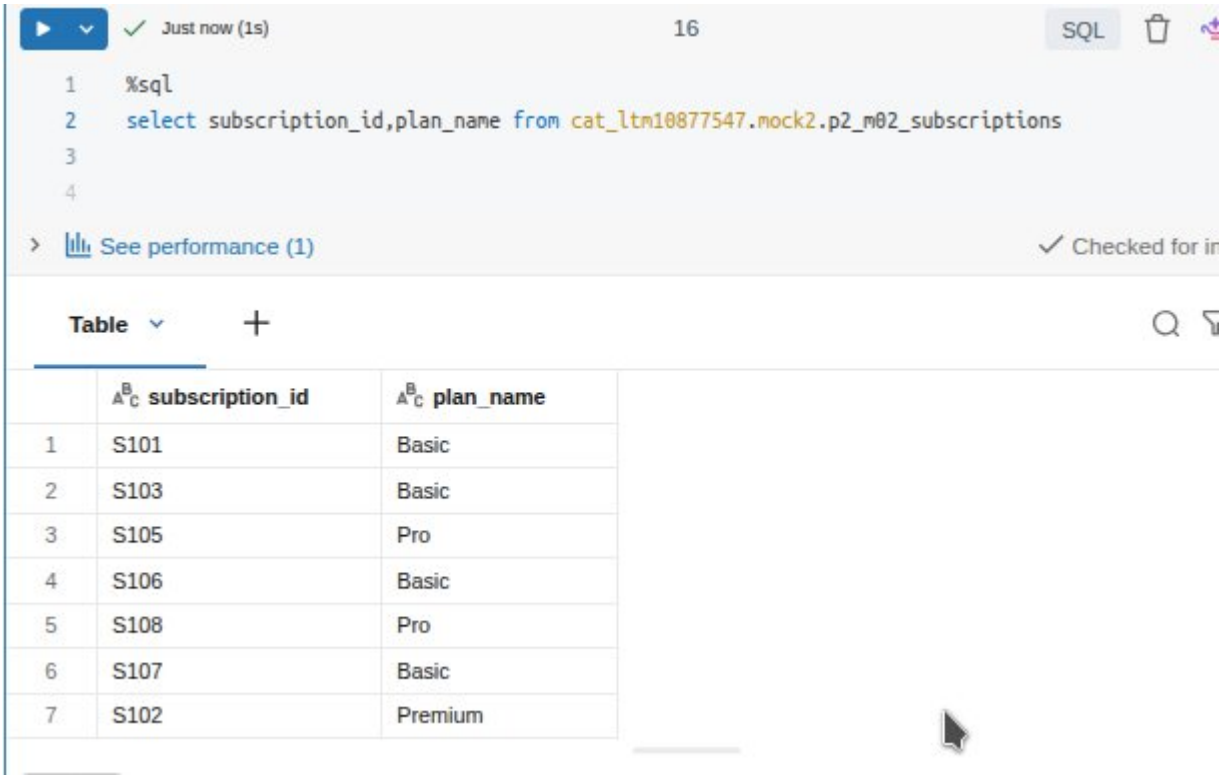
B. Results: 7-row final count, S104 absent and updated S102.

C. History: baseline and subsequent mutation versions clearly identifiable.

Use filenames such as M2_M02_T05_Code_A.png, M2_M02_T05_Result_A.png or M2_M02_T05_UI_A.png. Add _B / _C where required. These are naming examples, NOT a fixed screenshot count.

SCREENSHOT PLACEHOLDER - paste the required task evidence below; duplicate this holder for additional screenshots.

[Paste readable screenshot here]



	subscription_id	plan_name
1	S101	Basic
2	S103	Basic
3	S105	Pro
4	S106	Basic
5	S108	Pro
6	S107	Basic
7	S102	Premium

Additional screenshots (Code B / Result B / UI B, etc.) may be inserted after this placeholder.

Evidence reference / workspace object: _____

Expected final results

Task 1: 6 baseline records; CDF property true; actual baseline version captured.

Task 2: UPDATE 1; DELETE 1; INSERT 2; four incoming records.

Task 3: S102 Premium/999; S107 Basic/299; S108 Pro/599; snapshot after MERGE 8 rows.

Task 4: insert 2; delete 1; update_preimage 1; update_postimage 1; five CDF events.

Task 5: 7 subscriptions; S104 absent; S102 Premium/999; CDF counts as above.

Evaluation rubric and evidence map

Task	Rubric metric - evidenced by screenshots	Marks
1	Setup: volume location and both filenames.	2
1	Code/config: baseline managed Delta table and CDF enablement.	1
1	Results: CDF property true, initial row count 6 and baseline history version.	1
2	Code: typed change-file load with operation retained.	2
2	Result: grouped operation counts 1/1/2.	1
3	Code A/B if necessary: complete MERGE join and matched/unmatched actions.	2
3	Results: three named affected records and 8-row snapshot.	2
3	History: actual successful MERGE version, if shown separately.	1
4	Code: deletion logic referencing S104 and operation DELETE.	2
4	Code: CDF read with correctly recorded starting version and grouping by change type.	2
4	Result: four event categories with counts 2,1,1,1; history/version evidence if not already shown.	1
5	Code: final snapshot and row-count reconciliation.	1
5	Results: 7-row final count, S104 absent and updated S102.	1
5	History: baseline and subsequent mutation versions clearly identifiable.	1
TOTAL	No hidden checks or extra requirements	20

Submission requirements

Submit this matching template populated with readable taskwise screenshots. Provide 2 or 3 screenshots within a task when the implementation is long, or when code/configuration and results cannot both fit legibly. Show every assessed rubric metric, not just a success message. No fixed screenshot count, no scripts supplied in assessment instructions. Keep the executed workspace artifacts and history available. Do not rerun data-changing operations to recapture evidence.